Data science is the field that combines statistics, programming, and domain knowledge to extract insights and value from data.

Core Components:

Data Collection – Getting data from sources like databases, APIs, web scraping, or sensors.

Data Cleaning – Removing errors, dealing with missing values, normalizing formats.

Exploratory Data Analysis (EDA) – Using statistics and visualizations to understand the data.

Modeling – Using machine learning or statistical models to make predictions or find patterns.

Deployment – Making models available via apps, APIs, dashboards, etc.

Interpretation – Explaining the results in a meaningful way for decision-making.

Common Tools & Languages:

- Python (pandas, NumPy, scikit-learn, TensorFlow, etc.)
- R
- SQL
- Jupyter Notebooks
- Tableau / Power BI

Real-World Applications:

- Predicting customer behavior
- Recommender systems (like Netflix or Amazon)
- Fraud detection
- Health analytics
- Natural language processing